Self-organisation phenomena

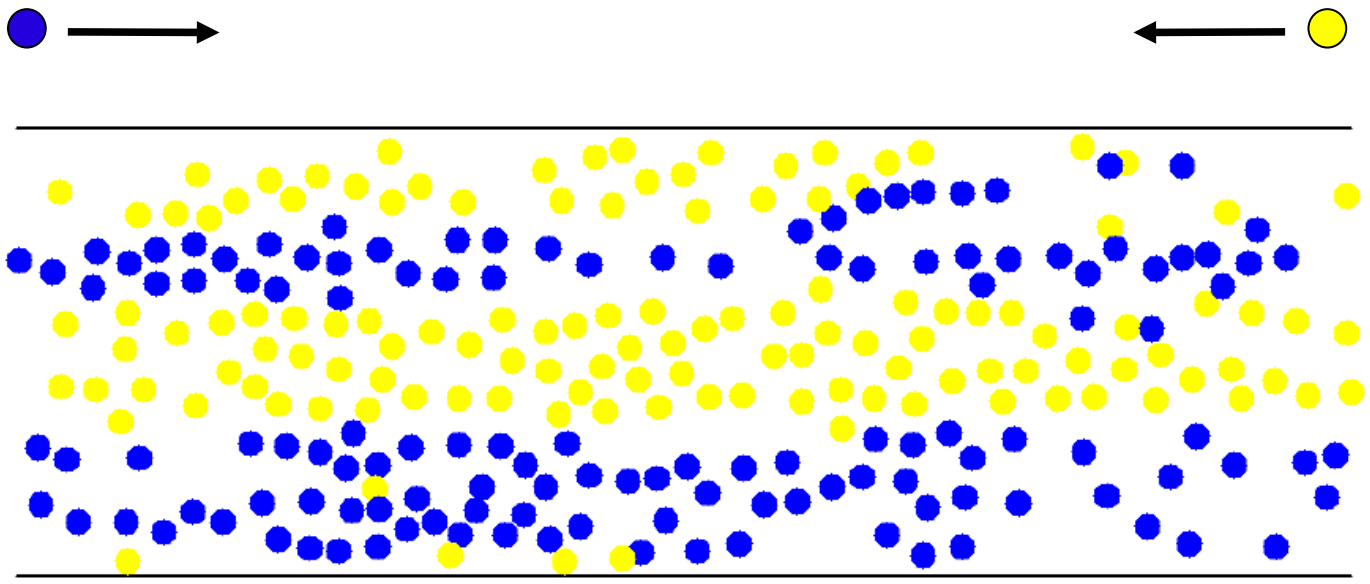
Now, when we have discussed the social force model on a microscopic perspective, we will also look at some scenarios from a macroscopic point of view.

More specifically, we will evaluate how well it can reproduce various self-organisation phenomena which have been observed in real crowds.

Properties of crowd dynamics; self-organisation phenomena

Let us now look at a number of emerging properties.

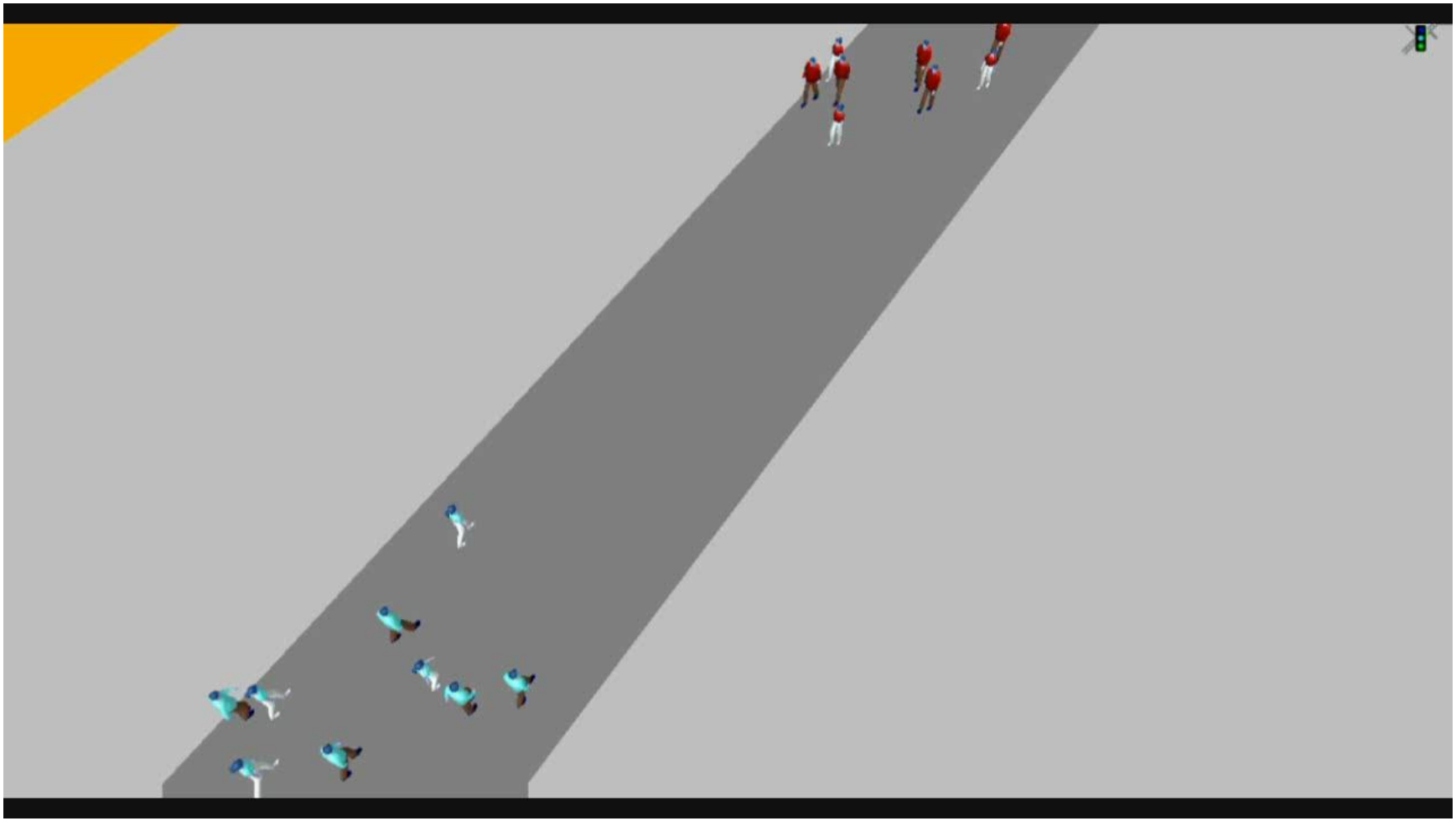
1) For bi-directional flows, *lane formation* of uniform walking directions will spontaneously emerge.



Empirical lane formation

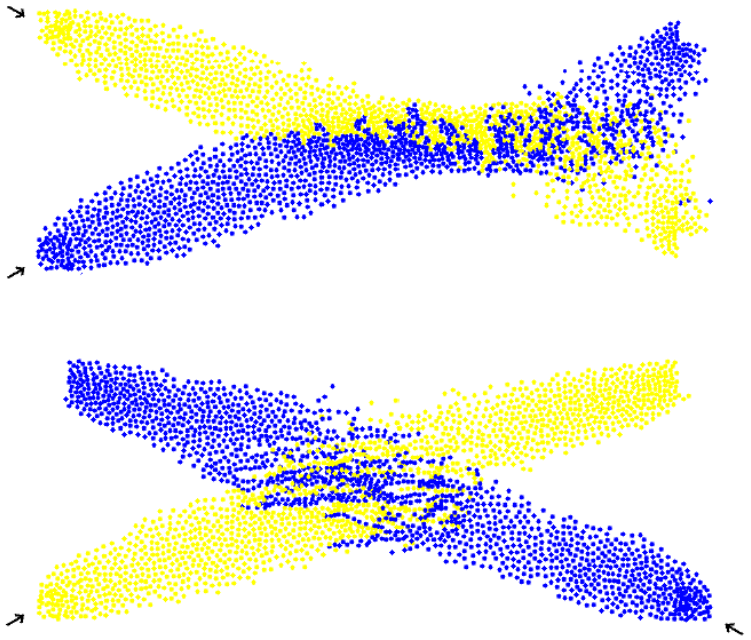


Simulated lane formation

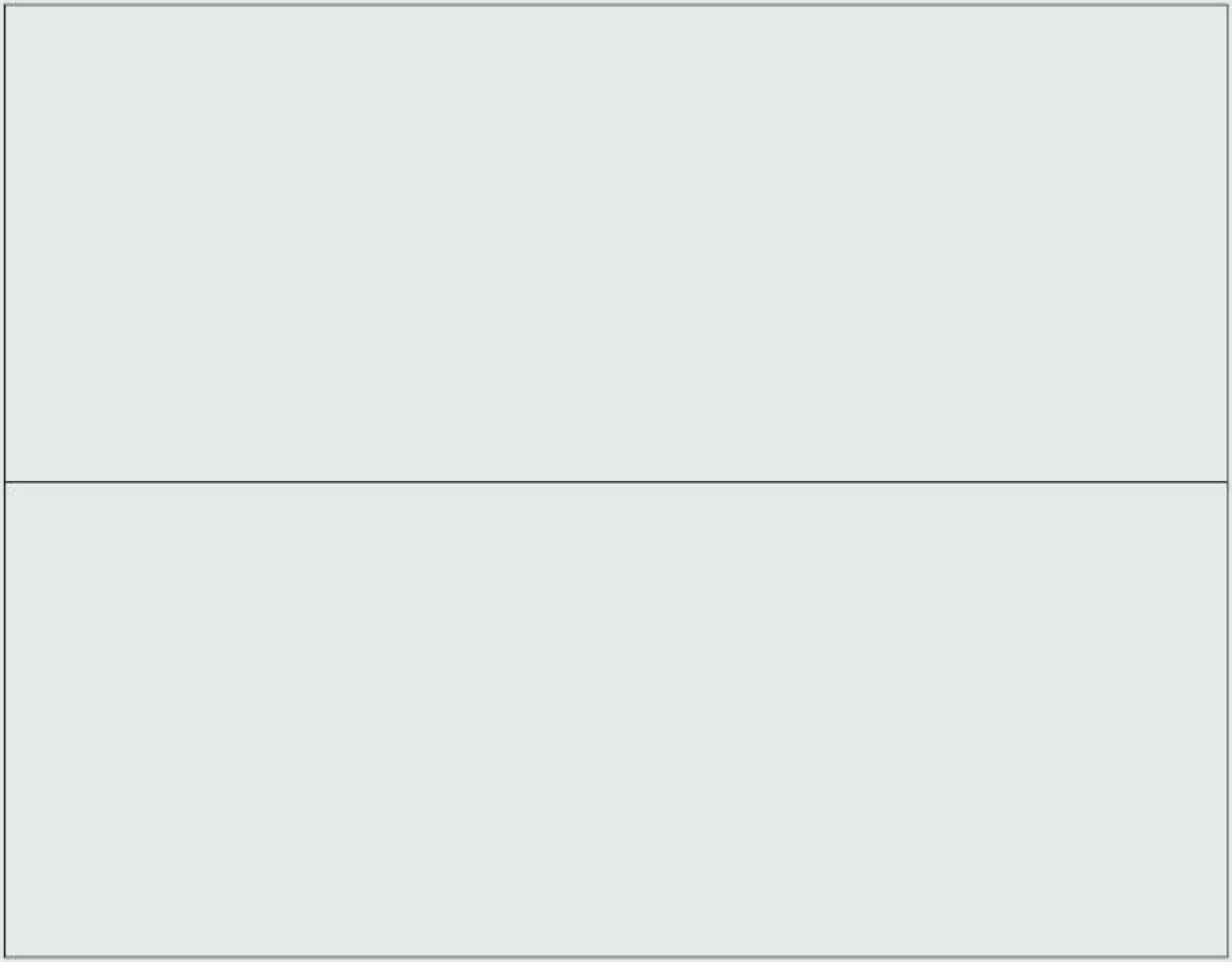


Self-organisation: Stripe Formation

If the two streams are intersecting with an angle different from 180 degrees, stripes will automatically form.

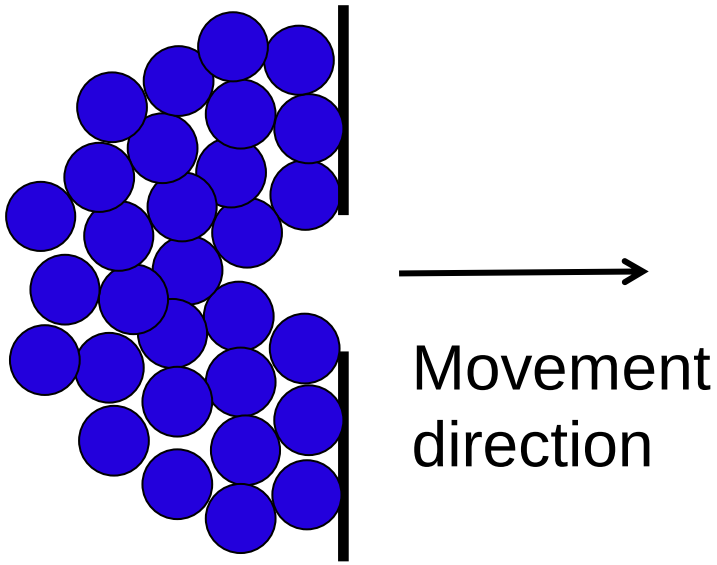


iterations: 0, time=0.000 s, pedestrians=0, satisfied=0

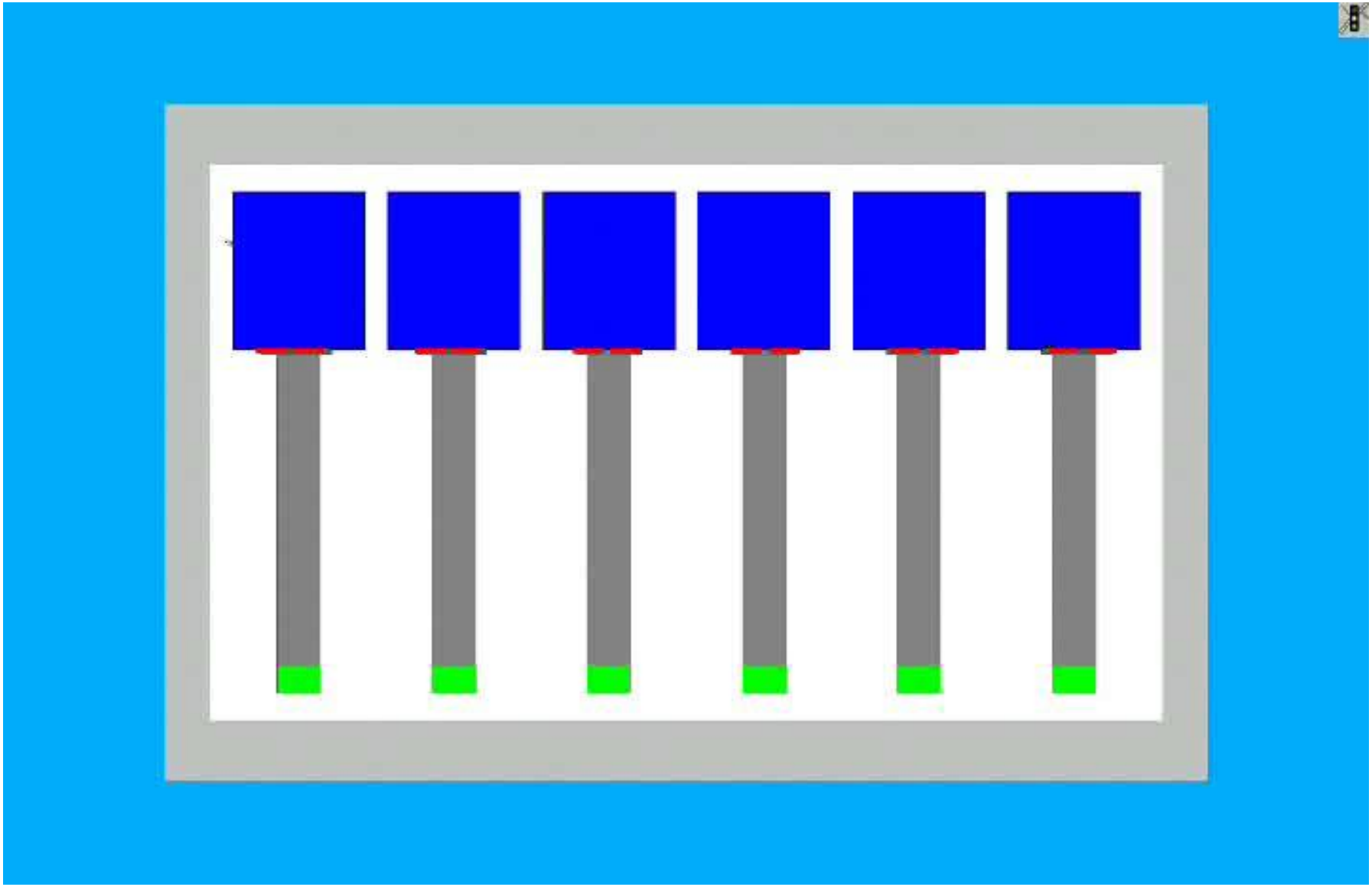


Clogging at bottlenecks

In evacuation situations, people may get stuck and clog up at bottleneck due to too much pushing and too little coordination.



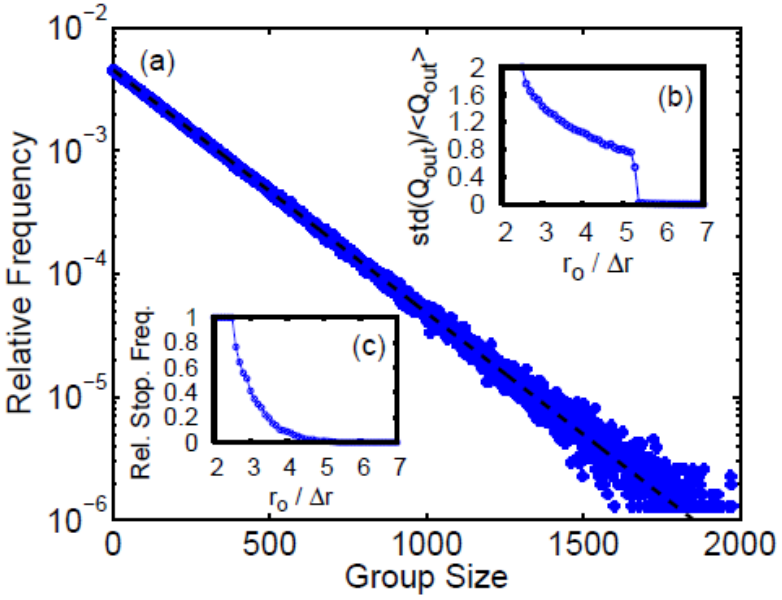




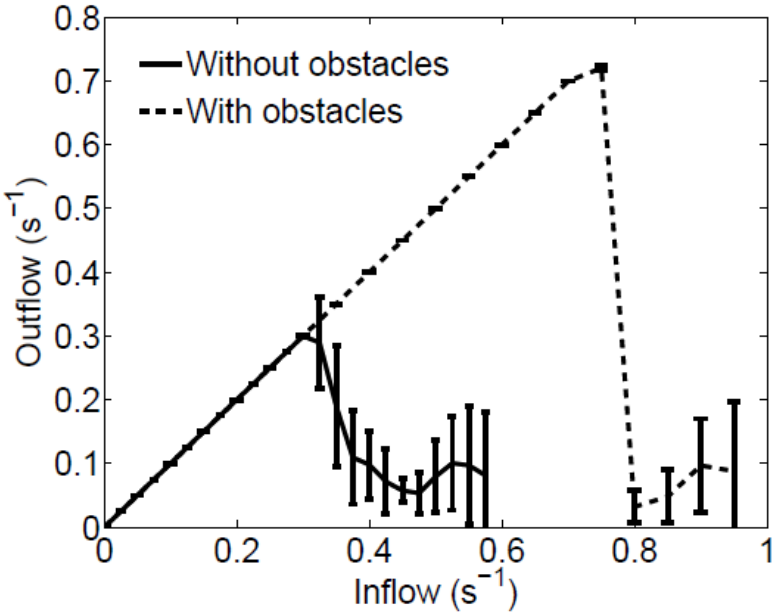
Clogging at bottlenecks

Problem: Inflow > Capacity => Sudden drop in flow

Solution: Adding obstacles can increase flow



Physical Review Letters **97**, 168001 (2006)



Transportation Science **39**(1), 1-24 (2005)

Model validation summary

Validation of models can be done in a number of ways:

- Visual: Use your engineering judgment (in CS: 'believability')
- Microscopic: Compare fit with empirical data for every individual agent, for every time step
- Macroscopic: Compare some aggregate quantity (such as the 'fundamental diagram')
- Phenomenological: List a number of 'stylized facts' of the real world and check how many of them a model can reproduce, and how well

Ideally, a combination of all of these will be used.